WEEK 4 PROGRESS REPORT

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

University Student-Support Case Agent

Group M Evening | Week Ending: 25th September 2026

Work Completed Against Weekly Objectives

Task	Status	Notes
Tool Catalogue and tool/API schemas	Completed	Defined two tools - retrieve_documents and create_support_case - with input schema, output schema, authorization and failure behaviour for each, plus a failure/authorization test matrix listing the minimum conditions to exercise.
Tool/function calling implementation	Completed	Wired both tools into the orchestration layer on a tool-enabled prompt path; retrieve_documents now replaces the automatic context injection used in earlier prompt versions.
Working demonstration of at least two tools	Completed	Demonstrated both tools against the live application on a dedicated feature branch.
Failure/authorization test evidence	Completed	Exercised both tools against missing parameters, unauthorized requests, unavailable services and unexpected responses; recorded results across 9 test conditions plus additional exploratory checks.
Updated architecture diagram	Completed	Revised the architecture diagram to show the tool-calling flow alongside the existing RAG path.
Human approval before higher-impact action	Completed	create_support_case stays off the model's tool list until the student explicitly agrees in-chat; admissions, grading and disciplinary requests are declined and never reach the tool, regardless of category.

Key Engineering Decisions

- Held the tool count to exactly two — retrieve_documents and create_support_case — matching the Project Charter's Minimum Proposal Statement, rather than adding extra tools for their own sake.
- Capped retrieve_documents at 3 chunks and dropped anything below 0.70 similarity, favouring precision over recall given the size of the corpus.
- Made student agreement for case creation (US-4) a session flag enforced by the orchestration layer rather than a model-supplied argument, since the model could otherwise mark a case “agreed” without the student having actually confirmed.
- Kept the student_identifier pattern and the category-to-queue mapping fully deterministic, so the model cannot choose, guess, or reformat a registration number or routing queue.
- Excluded admissions, grading and disciplinary matters from every category, including other_unclassified, and backed the prompt-level refusal with an orchestration-layer reject so the boundary does not depend on the model alone.

Failures, Challenges and Current Response

- Duplicate retrieve_documents calls: on at least two occasions a single webmail question produced two recorded tool calls instead of one. This does not violate the tool's documented schema or failure

behaviour, but it is a possible source of doubled retrieval cost. Response: flagged for investigation in Week 5 rather than fixed this week, so the root cause can be diagnosed properly.

- Silent duplicate case write under a 503 error: two high-demand errors shown to a student corresponded to two real case records being created about eight minutes apart, meaning the tool call had actually succeeded even though the student only saw a failure message. Follow-up controlled tests did not reproduce it consistently. Response: logged as a real, intermittent defect; recommend request-level logging in Week 5 to isolate whether the failure sits before or after the tool call completes.
- Unreliable 429 quota errors: the free-tier daily quota error sometimes recovered within about two minutes and sometimes did not recover for much longer, so the error's stated "daily" quota type is not a reliable signal of how long the block will last. Response: recommend a fixed short retry-with-backoff regardless of the stated cause; not yet implemented.
- issue_summary accepts near-empty input: minLength: 1 lets a student create a schema-valid case with no actionable content (e.g. "Something is wrong but I don't know how to explain it"). Response: flagged as a design limitation rather than fixed immediately, to avoid unplanned scope this week; tightening it is carried into next week's plan.

Links and Evidence

GitHub repository: https://github.com/SumayyaN/BSE41014_capstone_group_m_eve

ClickUp board (Team space, Week 4):

https://app.clickup.com/1200410000000374/v/b/li/1200410000022716

Individual Contribution Summary

Member	Task Owned	Evidence
Nakaye Hildah	Tool Catalogue and API Schemas; Week 4 progress report	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/docs/requirements/T
Aryampa Joshua	Working demonstration of at least two tools	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/tree/feature/week4-working-de
Mushabe Moses	Updated architecture diagram	
Namata Sumayya	Failure/authorization test evidence	

Plan for Next Week (Week 5 – Agent Architecture and Bounded Autonomy)

- Define one task that genuinely benefits from multi-step decision making.
- Design the Sense/Context → Plan/Decide → Act/Tool → Observe → Stop/Re-plan loop.
- Set maximum iterations, approved tools, stop conditions and human hand-off/approval conditions.
- Implement the workflow using direct orchestration or a framework.
- Capture at least three execution traces, including one failure/recovery case.